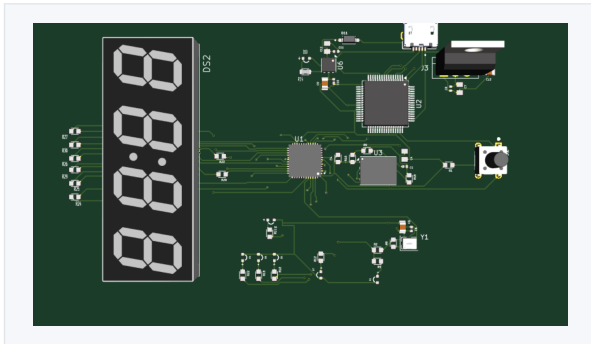


PCB Design Work Samples

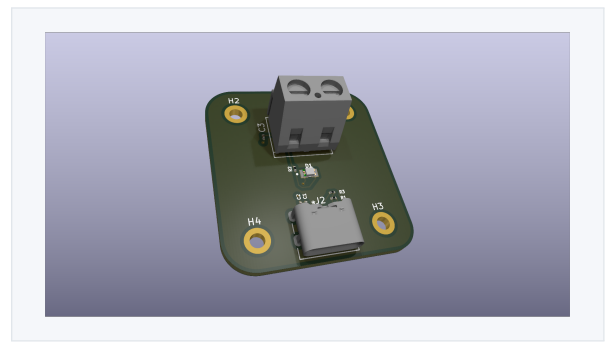
Moiz Zaheer Malik

Electronic Engineering | Hochschule Hamm Lippstadt

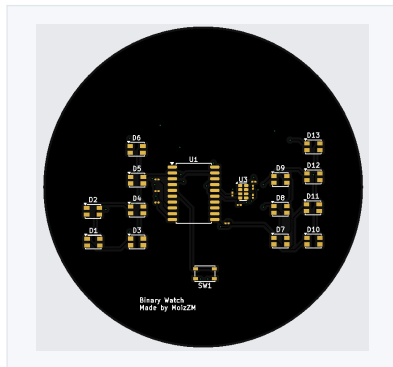
KiCad schematic design, PCB layout, component placement, routing, ERC, DRC, Gerber files, and 3D PCB views.



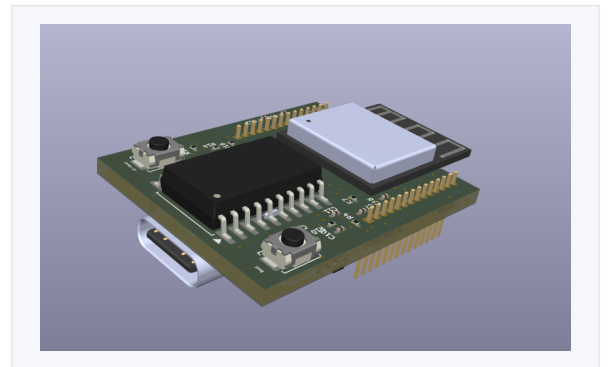
Custom FPGA Development Board



USB C Power Supply PCB



Binary Watch PCB



ATtiny Microcontroller Board

Focus areas

schematic design | compact PCB layout | hardware planning | manufacturing files |
practical board development

Portfolio document

0.1 Custom FPGA Development Board

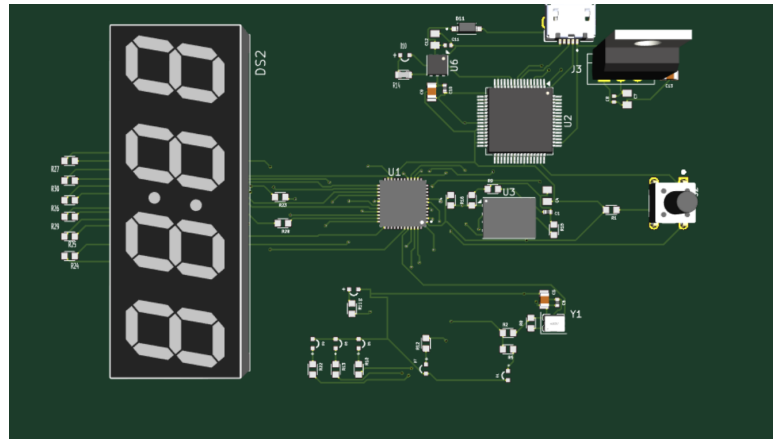


Figure 1: 3D view of the custom FPGA development board showing the FPGA, USB interface, SPI Flash, regulators, reset button, status LED, and 4 digit 7 segment display.

Context: Hardware Engineering Lab project

Tool used: KiCad

Short description: This project focused on the PCB design of a custom FPGA development board based on the Lattice iCE40UP5K FPGA. The board included USB programming and UART, SPI Flash, voltage regulation, reset control, LEDs, and a 4 digit 7 segment display.

My contribution:

- Led the PCB design part of the project.
- Created the schematic and organized it using hierarchical sheets.
- Selected the main components for power, USB, flash memory, clock, reset, LEDs, and display.
- Designed the PCB layout in KiCad based on signal flow and power flow.
- Manually routed important signals such as USB, SPI, and clock lines.
- Checked the design using ERC and DRC before generating Gerber and drill files.

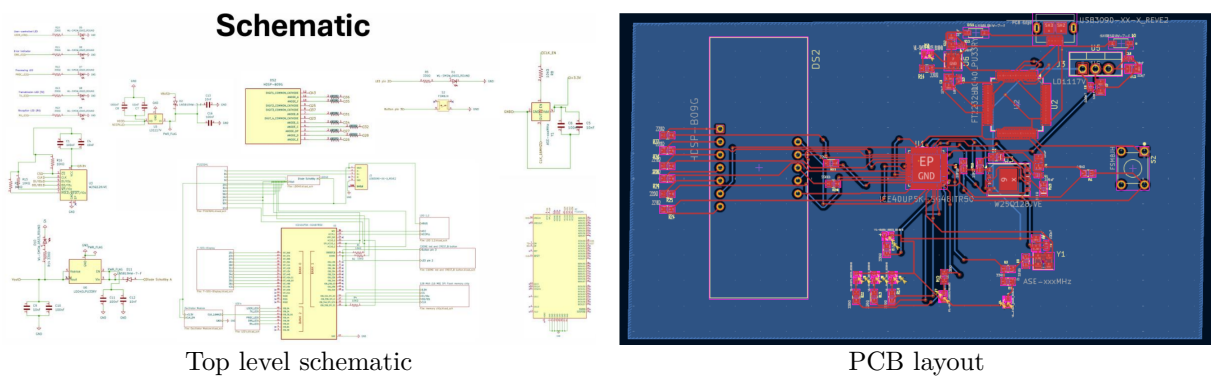


Figure 2: Schematic and PCB layout of the custom FPGA development board designed in KiCad.

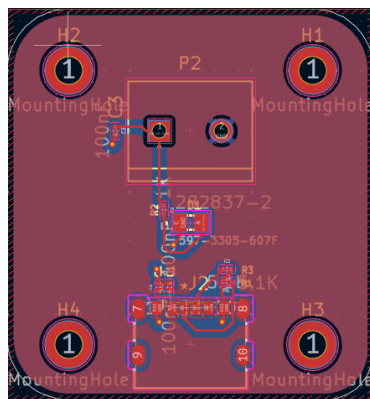
Result: This project gave me practical experience in schematic design, PCB layout, component selection, signal routing, design checks, and manufacturing file preparation.

0.2 USB C Power Supply PCB

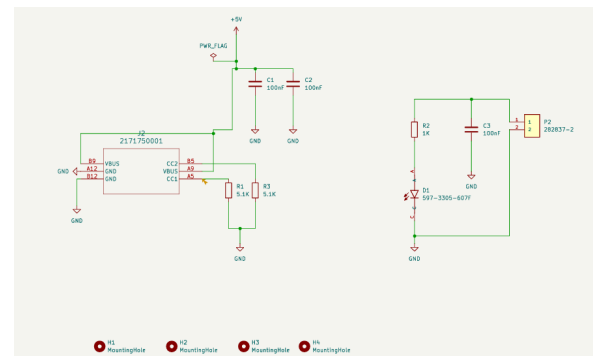
Context: Personal PCB design project

Tool used: KiCad

Short description: This project focused on designing a small USB C power supply PCB. The board provides 5 V output through a USB C connector and includes CC resistors, a decoupling capacitor, a power LED, an output header, and mounting holes.



PCB layout

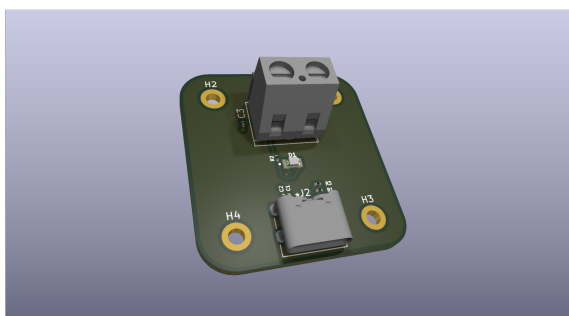


Schematic

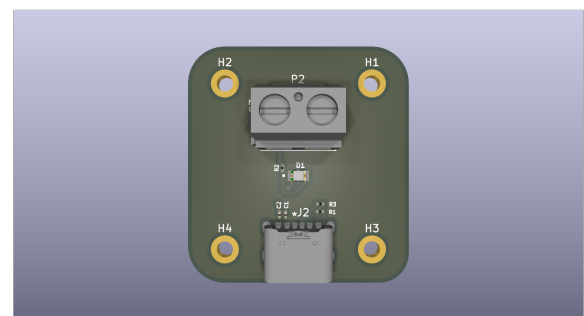
Figure 3: PCB layout and schematic of the USB C power supply board designed in KiCad.

My contribution:

- Designed the schematic and PCB layout in KiCad.
- Added CC1 and CC2 resistors for USB C configuration.
- Added a decoupling capacitor, power LED, output header, and mounting holes.
- Checked the schematic and layout using ERC and DRC.
- Generated Gerber, Drill, BOM, and Pick and Place files.



3D diagonal view



3D top view

Figure 4: 3D views of the USB C power supply PCB showing the USB C connector, output header, power LED, and compact board shape.

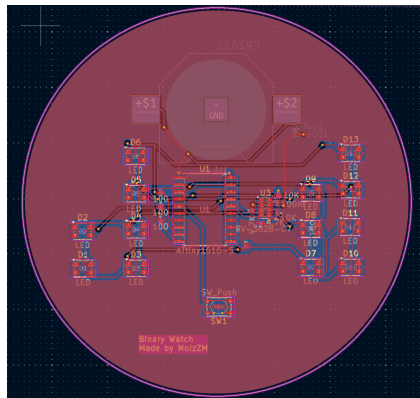
Result: This project gave me practical experience in small PCB design, USB C power connection, component placement, design checks, and manufacturing file preparation.

0.3 Binary Watch PCB

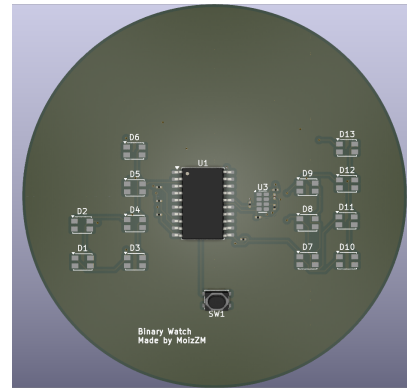
Context: Personal PCB design project

Tool used: KiCad

Short description: This project focused on designing a compact binary watch PCB. The board includes LEDs, microcontroller based control, battery related circuitry, and a compact PCB layout.



PCB layout

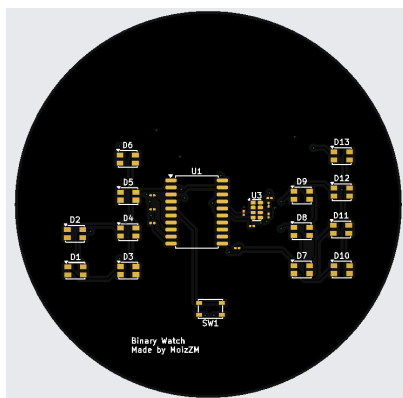


3D top view

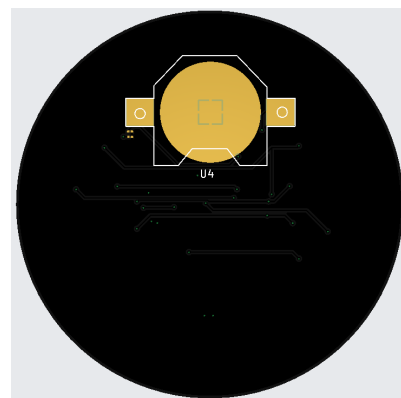
Figure 5: PCB layout and 3D top view of the binary watch PCB designed in KiCad.

My contribution:

- Designed the schematic and PCB layout in KiCad.
- Placed LEDs and control components based on the watch layout.
- Worked on compact component placement for a small board shape.
- Checked the schematic and PCB layout using ERC and DRC.
- Prepared the design for manufacturing.



Manufactured PCB front side



Manufactured PCB back side

Figure 6: Manufactured binary watch PCB received from JLCPCB, showing the front and back side of the board.

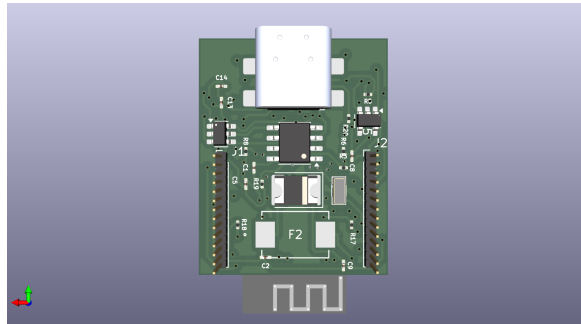
Result: This project gave me practical experience in compact PCB layout, LED placement, wearable style board design, and preparing a small electronics design for manufacturing.

0.4 ATtiny Microcontroller Board

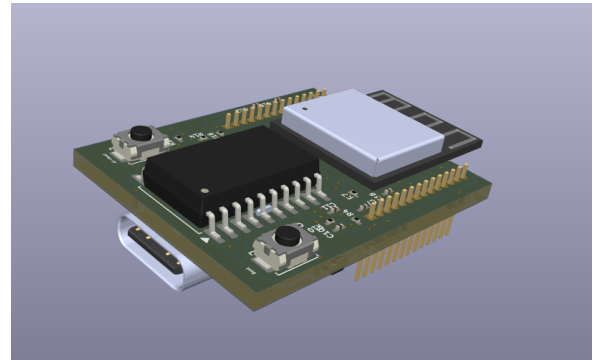
Context: Personal PCB design project

Tool used: KiCad

Short description: This is an ongoing custom microcontroller board project based on an ATtiny MCU. The board includes USB C input, voltage regulation, reset and clock support, programming connections, and external headers.

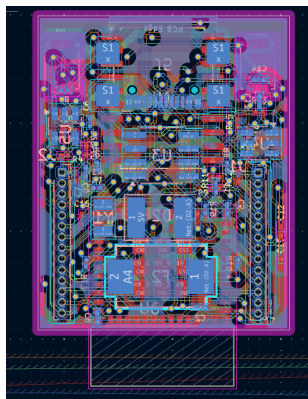


3D top view

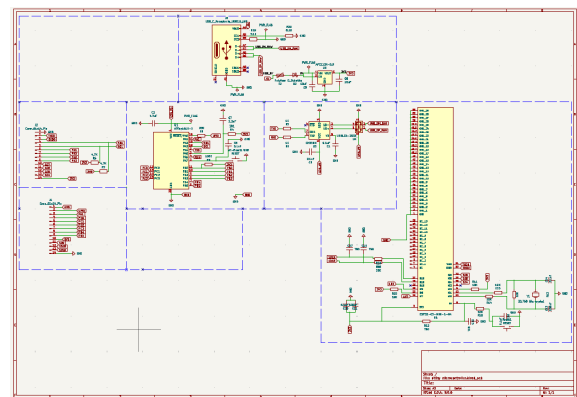


3D diagonal side view

Figure 7: 3D views of the ongoing ATtiny microcontroller board designed in KiCad.



PCB layout



Schematic

Figure 8: PCB layout and schematic of the ATtiny microcontroller board.

My contribution:

- Designed the schematic structure in KiCad.
- Added the ATtiny MCU and supporting circuits.
- Added USB C input, voltage regulation, reset, clock, and connector sections.
- Designed the PCB layout and placed the main components.
- Prepared the project for final checks and manufacturing files.

Current status: This project is still in progress. The schematic and layout are prepared, but final design checks and manufacturing files still need to be completed.

Result so far: This project shows my practice with custom MCU board design, schematic planning, PCB layout, and microcontroller support circuits.